

VEDANT BHADRI

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PERSONAL PROFILE

A quantitative and analytically-minded Computer and Information Sciences master's student, with hands-on experience building data-driven systems for live financial markets, risk-aware trading logic, and large-scale optimisation problems. A fast learner across new tools, systems, and technical domains, with a genuine curiosity for how data and analytics drive real-world decision-making - now looking to bring this problem-solving mindset to Aotearoa New Zealand's energy market.

WORK EXPERIENCE

Teaching Assistant

Feb 2026 – Present

Auckland University of Technology, Auckland, New Zealand

Course: COMP517 Data Analysis | COMP501 Computing Technology in Society

- Support undergraduate students through weekly lab sessions, guiding experiments, data analysis, and problem-solving under time pressure.
- Grade assignments and lab reports, delivering clear, constructive feedback to drive better student outcomes.
- Build communication and adaptability by working effectively with people across a wide range of technical skill levels.

Software Developer Intern

Jun 2024 – Sep 2024

Wynby Technologies, Bangalore, India

- Built backend infrastructure in Python for a live algorithmic trading system, developing REST APIs and SQL-based data pipelines to process real-time market data.
- Designed trading logic - including trailing stop-loss mechanisms and pattern-recognition algorithms - to support automated strategy execution and manage downside risk.
- Collaborated cross-functionally with traders and engineers to deploy backend functionality, improving system reliability and performance under live, fast-moving market conditions.

ACADEMIC PROJECTS

Smart-City Delivery Route Optimisation Using Genetic Algorithms · Nature Inspired Computing, AUT

- Designed an Enhanced Genetic Algorithm (EGA) for the Capacitated Vehicle Routing Problem with Time Windows under realistic smart-city constraints, incorporating adaptive mutation and 2-Opt local search.
- Achieved a 32.7% reduction in total travel distance and 25% improvement in fitness score vs. baseline GA, with 100% on-time delivery - demonstrating measurable impact of algorithmic optimisation on operational efficiency.

Non-Invasive Blood Glucose Estimation from PPG: Deep Learning · Master's Thesis · Ongoing

- Iteratively developed and evaluated deep learning models on physiological signal data, improving accuracy through systematic experimentation and feature engineering - reflecting a structured, analytical approach to open-ended technical problems.
- Integrated demographic and physiological context features into the modelling pipeline, with current work extending toward real-time hardware deployment.

EEG-Based ADHD Classification: ML & Spiking Neural Networks · Neuroinformatics, AUT

- Applied and compared multiple machine learning models (SVM, MLP, Evolving Classification Function) to classify EEG signal data, achieving up to 67.6% accuracy - demonstrating rigorous, iterative model evaluation on complex, noisy real-world data.
- Extended the analysis using a Spiking Neural Network framework, encoding EEG data into spike trains and mapping it spatially in a 3D brain model to improve classification performance.

Semantic Image Retrieval Using CNN Embeddings & FAISS · Artificial Intelligence, AUT

- Built a content-based image retrieval pipeline using a pretrained ResNet-50 for feature extraction and FAISS for scalable nearest-neighbour search, validating design choices through a rigorous comparative evaluation.

EDUCATION

Master of Computer and Information Sciences

Auckland University of Technology, Auckland

Jul 2025 – Present

Bachelor of Engineering in Information Science and Engineering

KLS Gogte Institute of Technology, India

Nov 2020 – Jul 2024

SKILLS

Programming Languages: Python, SQL

Libraries & Frameworks: NumPy, scikit-learn, PyTorch, TensorFlow/Keras, FAISS

Software & Tools: MySQL, Git, Postman, VS Code, Jupyter Notebook, NeuCom/NeuCube, Raspberry Pi

Technical Concepts: Data Analytics, Automation, Machine Learning, Deep Learning, Signal Processing, Computer Vision, NLP, REST APIs, Database Management, Algorithm Design

Soft Skills: Problem Solving, Leadership, Communication, Teamwork, Adaptability

RELEVANT COURSEWORK

Data Science, Artificial Intelligence, Deep Learning, Machine Learning, Data Mining, Python Programming, Neuroinformatics, Nature Inspired Computing, Soft Computing, Robotic Process Automation, Database Management, Computer Networks, Design and Analysis of Algorithms, Operating Systems.

LEADERSHIP AND VOLUNTEERING EXPERIENCE

General Executive

AUT Off-Grid Club

Apr 2026 – Present

- Assisted in organising hiking, camping, and outdoor adventure events; coordinated participant registrations and grouped students for smooth event management.

Volunteer Retail Assistant

New Zealand Red Cross

Dec 2025 – Feb 2026

- Operated checkout, handled cash and EFTPOS transactions, managed donated stock, and maintained shop presentation, demonstrating reliability and team collaboration.

Volunteer

Rotaract Club of GIT

Jan 2021 – Jul 2024

- Participated in community service projects and fundraising initiatives to create positive community impact.